

Gruebler–Kutzbach Mobility Formula for Planar Mechanisms

$$M = 3(n - 1) - 2 j_1 - j_2$$

n = number of links

Include ground link (link 1)

Each free link = 3 DOF

j₁ = full joints

Revolute (R), Prismatic (P)

Each removes 2 DOF

j₂ = half joints

Cam pair, Gear pair

Each removes 1 DOF

Compound Joint Rule

If m links share one joint location → count it as (m - 1) full joints

Example: 3 links at one pin = 2 full joints (j₁ += 2, not 1)